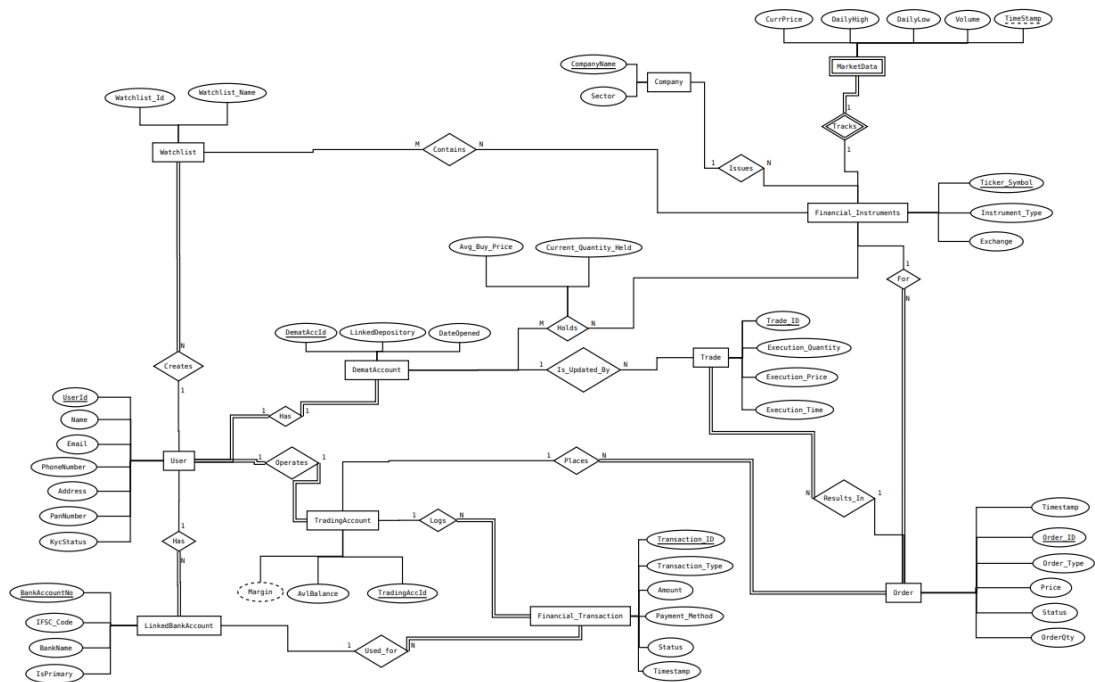
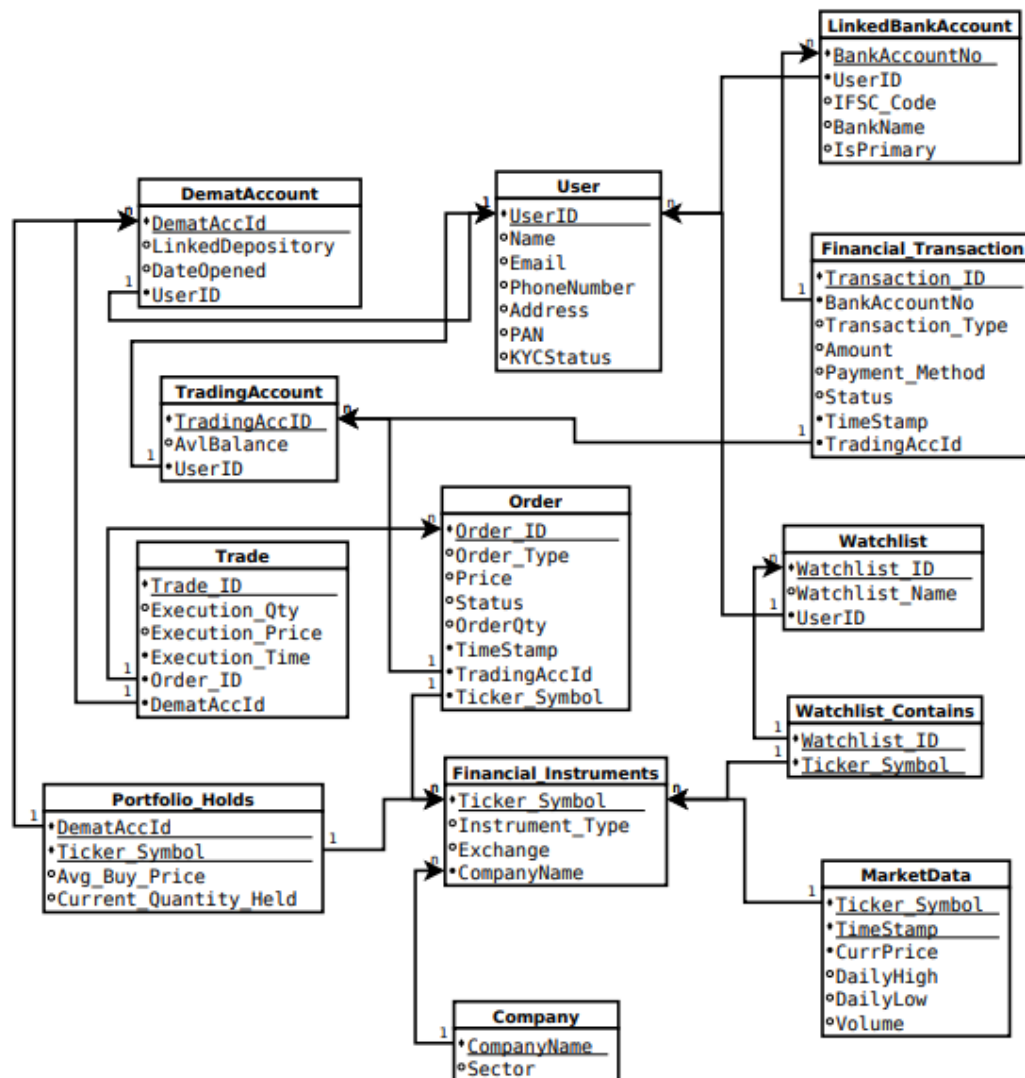


ER



Relational Schema



Proof that relations are in BCNF

- To identify whether a database is in BCNF, we first determine all the functional dependencies present in each relation.
- A relation is in Boyce-Codd Normal Form (BCNF) if, for every non-trivial functional dependency $X \rightarrow Y$, the determinant X is a superkey. After identifying primary and candidate keys, each dependency is checked against this condition.

- For example, in the given schema, in the **AppUser** relation we have $\text{UserId} \rightarrow \text{Name}, \text{Email}, \text{PhoneNumber}, \text{etc.}$, where **UserId** is the primary key, so it satisfies BCNF.
- Additionally, dependencies like $\text{Email} \rightarrow \text{UserId}$ and $\text{PhoneNumber} \rightarrow \text{UserId}$ also hold, but since **Email** and **PhoneNumber** are defined as **UNIQUE**, they act as candidate keys, and thus still satisfy BCNF. Similarly, in the **MarketData** relation, $(\text{Ticker_Symbol}, \text{TimeStamp}) \rightarrow \text{DailyHigh}, \text{DailyLow}, \text{Volume}$, and the determinant is the composite primary key, so it also satisfies BCNF.
- Since in all relations the determinants are superkeys or candidate keys, **the schema is already in BCNF**. If any dependency had a determinant that was not a superkey, we would decompose the relation into smaller relations to achieve BCNF while ensuring a lossless join.